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## How the Two Models Connect

The economic channels of the OG-Core  $\times$  CLEWS soft link —  
and how each is represented in the model

numbers: the Philippine reference run · `ogclews-link` · July 6, 2026



## **The coupling problem**

## Two models, complementary blind spots

### CLEWS / OSeMOSYS

a least-cost optimiser: chooses what to build and run so that energy demand is met at the lowest discounted system cost

- **rich:** technology, season-by-season operation, emissions, land & water
- **blind:** demand is *fixed* — no price response, no households, no government budget, no distribution

Each is detailed exactly where the other is blank — so what usefully crosses the seam differs by direction:

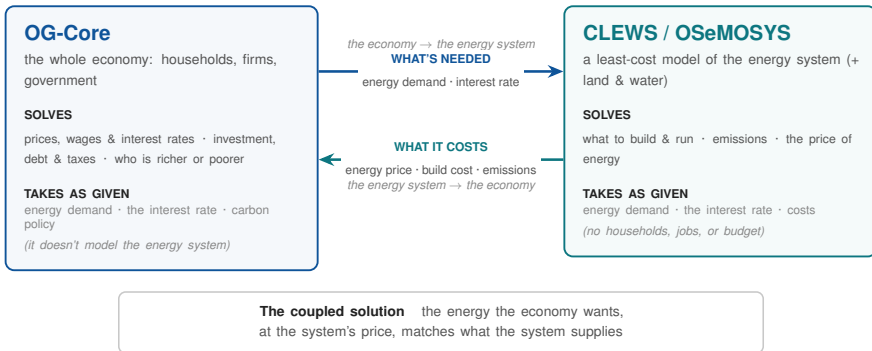
**quantities** flow from the economy, **prices and physical outcomes** flow from the energy system.

### OG-Core

overlapping-generations general equilibrium: households by age  $S \times$  lifetime income  $J$ ,  $M$  industries, a government that closes its budget each period

- **rich:** behaviour, saving, fiscal policy, incidence, demography
- **blind:** no energy technology — energy is not a priced input to production

# The structural seam



*Each model is detailed exactly where the other is blank.*

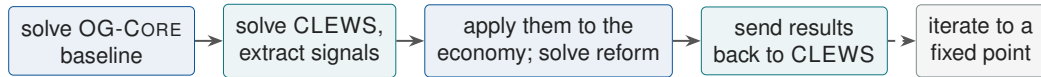
Quantities flow right (energy demand, the interest rate); prices and physical outcomes flow left (the energy price, build cost, emissions). The models stay separate and pass a few auditable signals.

## Why a *staged soft* link, not one big model

### A hard link is ruled out structurally, not practically:

- OG-CORE solves an overlapping-generations *transition path* period by period; OSEMOSYS is one perfect-foresight LP over all years at once
- OG-CORE exposes no scalar welfare into which the LP could be folded; no shock can be injected mid-solve

### So the models stay separate and the link runs in stages:



Validated today: the single pass (solid). The iterated loop (dashed) is designed — the return signals are already produced in the right form — but not yet wired.



**Signals, wedges, and the price that carries the load**

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## The vocabulary: **signal** → **wedge** → **channel**

**Signal** a quantity carried out of a *solved* CLEWS scenario: the marginal cost of a fuel (what one more unit would cost to supply), the capital a build-out requires, emissions, the cost implied by an emissions cap (an implied carbon price).

## The vocabulary: signal → wedge → channel

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**Wedge** the OG-CORE parameter that expresses it: the consumption-tax vector  $\tau^c$ , an industry productivity  $Z_m$ , the public-investment share  $\alpha_I$ , mortality  $\rho(s, t)$ , effective labour  $e(t, s, j)$ . (“Wedge” in the public-finance sense — the simplest channel literally inserts a tax wedge between producer and consumer price.)



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**Channel** a named, guard-railed map from one signal to one wedge — plus the validation that fires when the mapping is large or ill-posed.

Each channel is one small, testable mapping — and nothing country-specific is built in: which industry is electricity, and how many industries there are, is read from the country’s own model at run time.

## The load-bearing signal: the system's own marginal cost

The energy model must balance supply and demand for every fuel, in every year and season. Ask how much total system cost would rise if one more unit had to be supplied, and you get that fuel's **marginal cost** — under the usual competitive reading, *the price that quantity would fetch in a market*.

### 1. It closes an elasticity gap

Inside the energy model, demand does *not* respond to price. Handing this price to OG-CORE households — whose energy use responds to price but cannot fall below a subsistence floor — supplies the demand response the optimiser lacks.

### 2. It reveals incidence

A single “cost” is silent on who bears it.  $S \times J$  households distribute the burden; because energy is a partial necessity, an energy-price change is *regressive* — invisible to a single-agent model.

## When are the two models consistent? (the target)

$D^* = D_{\text{OG}}(\pi^*)$  and  $\pi^* = \pi_{\text{C}}(D^*)$ : the economy demands  $D^*$  at price  $\pi^*$ , and  $\pi^*$  is the system's marginal cost of delivering  $D^*$ . A single pass uses the price at *baseline* demand; iterating to the fixed point closes the loop.

## Three routes for one energy cost

OG-CORE production has no energy input — an energy-cost signal must enter through one of three routes:

- A demand-side tax wedge** —  $1 + \tau_{t,i_e}^{c,r} = g_t (1 + \tau_{t,i_e}^{c,b})$ ,  $g_t$  the reform/baseline price ratio. Reuses existing structure, delivers incidence; but household-only, and the “revenue” is phantom. Empirically  $\approx$  output-neutral.
- B supply-side productivity haircut** — each industry’s cost rises by its direct energy share plus what suppliers pass through ( $\hat{c} = (I - A^\top)^{-1} \theta \hat{p}_E$ ), applied as per-industry productivity reductions. Right sign, materially contractionary; but firms cannot substitute away and the magnitude is a proxy.
- C energy as a priced production input** — firms buy energy alongside capital and labour, so the energy price feeds firm costs directly and the loop closes on quantities too. The structural ideal; requires an extension of OG-CORE (unbuilt).

**The route is a first-order modelling choice:** routes A and B carry the *same* physical cost to opposite macro verdicts — roughly neutral vs. sharply contractionary. The structurally faithful route (C) is therefore the target, not a refinement.



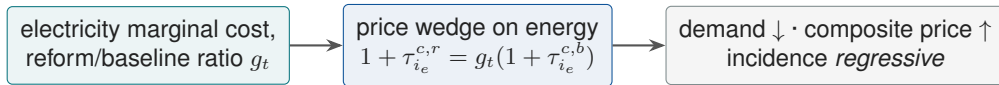
## **The eight channels**

## The map

| Channel              | Dir. | CLEWS signal                                | OG-CORE object                                 | First-order effect                         |
|----------------------|------|---------------------------------------------|------------------------------------------------|--------------------------------------------|
| Energy price         | c→o  | electricity marginal cost (reform/baseline) | price wedge ( $\tau^c$ ) on the energy good    | demand ↓; regressive incidence             |
| Carbon price         | pol  | carbon price (\$/tCO <sub>2</sub> )         | energy-wedge add-on + emissions penalty        | demand ↓; emissions ↓                      |
| Public investment    | c→o  | grid & transmission build cost              | public-investment share → public capital       | public capital ↑; crowding-out, debt       |
| Capital intensity    | c→o  | power-fleet capital-cost share              | energy industry's capital share ( $\gamma_m$ ) | energy price ↓; energy $K$ ↓               |
| Investment incentive | pol  | tax-credit rate (policy response)           | cost of capital (investment tax credit)        | capital reallocates into energy            |
| Health               | c→o  | PM <sub>2.5</sub> emissions change          | survival rates + worker productivity           | mortality $\approx 0$ ; morbidity ↑ labour |
| Discount rate        | o→c  | —                                           | economy's return → planning discount rate      | shifts the build mix (via the loop)        |
| Energy demand        | o→c  | —                                           | economy's activity → energy demand             | more capacity (via the loop)               |

four CLEWS→OG-CORE · two policy levers applied once to both sides · two OG-CORE→CLEWS

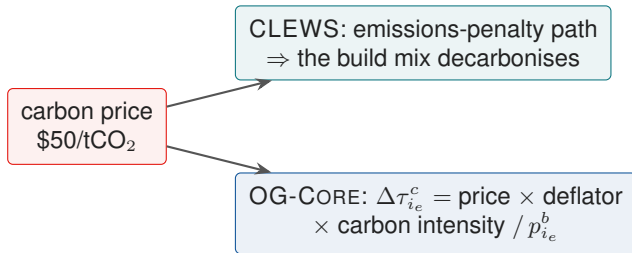
## Energy price — the system's marginal cost reaches household budgets



- households face  $(1 + \tau_{i_e}^c) p_{i_e}$ ; energy quantity responds — the price response CLEWS lacks
- with a subsistence floor ( $c_{\min} > 0$ ), poorer households spend a larger energy share  $\Rightarrow$  they bear more of a price change, even when the revenue is returned to them (the recycled variant:  $C_{ss} = -0.087\%$ )
- the price handed over is the scenario's *real* electricity price — a curated cost-of-electricity index where one exists, otherwise the solved system's marginal cost — and for the Philippine pair it is **near-flat**, so the standalone effect is  $Y_{ss} = -0.0005\%$ : honestly transmitted, the data does little
- the earlier  $\pm 20\%$  stress runs proved the *transmission*; they are retained only for the route comparison

maturity: route A, on the real price; the coupled run transmits the same price through the composite (later).

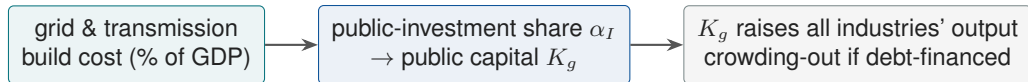
## Carbon price — one price, two sides, counted once



- one chosen price enters *both* models along the same path — never read back a second time from the energy model's own implied carbon cost
- OG-CORE firms buy no energy, so only *household* energy carbon is priced on the macro side; industrial carbon is priced in CLEWS
- revenue is recyclable (transfers); standalone it is the clearest contraction:  $Y_{ss} = -0.040\%$

the *level* rides on the uncalibrated unit/deflator bridge — mechanics validated, magnitude illustrative.

## Public investment — the grid on the government's books



- this is where crowding-out *legitimately* lives: debt-financed spending competes for the savings pool, raises  $r$ , displaces private investment
- public-grid capital and private-generation capital touch different objects — coupling both is complementary, not double counting
- Philippine scenario: the *extra* public grid spending is  $\approx 0$  (the transition happens on the generation side)  $\Rightarrow$  the channel is inert here ( $Y_{ss} = +0.000\%$ ); its mechanics are validated and scale with any non-zero spending



## The capital pair I: capital intensity — a worked example

**Signal:** reform/baseline ratio of the power fleet's capital-cost share  $\Rightarrow$  scales the energy industry's capital exponent  $\gamma_m$  (here +12.2%).

**The model's own first-order condition** (Cobb–Douglas,  $\varepsilon = 1$ ): the user cost  $\rho_m$  and taxes cancel in the reform/baseline ratio, leaving an exact identity —

$$K_m = \frac{\gamma_m p_m Y_m}{\rho_m} \quad \Rightarrow \quad \hat{K}_E = \hat{\gamma}_E \cdot \hat{p}_E \cdot \hat{Y}_E$$

|                               | $\hat{\gamma}_E$ | $\hat{p}_E$ | $\hat{Y}_E$ | $\hat{K}_E$ (obs.) | identity |
|-------------------------------|------------------|-------------|-------------|--------------------|----------|
| energy industry ( $m_e = 2$ ) | +12.2%           | −42.8%      | +20.9%      | −22.4%             | −22.4%   |

- the higher exponent cuts the sector's unit cost  $\rightarrow$  under zero profit the electricity price **collapses** −42.8%
- demand is elastic enough that output **rises** +20.9% — but revenue  $p_m Y_m$  shrinks by more than  $\hat{\gamma}$  and  $\hat{Y}$  add back  $\Rightarrow$  capital **falls** −22.4% (identity closes to 0.03%)

Raising the capital *share* sheds capital: this is a factor-share and energy-price lever, *not* a capital draw-in.

## The capital pair II: the investment incentive — the actual draw-in lever

The capital-*attraction* force is a structurally different instrument: an **investment tax credit** on the energy industry.

$$\rho_m = \frac{r + \delta - \tau_m^b \delta \tau_m^{\tau} - \tau_m^{\text{inv}} \delta}{1 - \tau_m^b}$$

the ITC enters the *user-cost denominator*, where  $\gamma_m$  is absent  $\Rightarrow$  capital demand shifts *out* at given  $r$

### the capital share $\gamma_m$ (capital intensity)

moves factor *demand* through the production exponent; absent from  $\rho_m$ . Result: cheaper electricity, *less* energy capital.

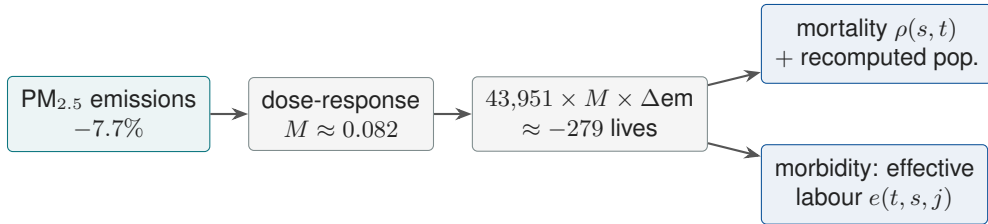
### the tax credit (investment incentive)

moves the *cost of capital*; absent from the exponents. Result: capital reallocates *into* energy. (A defined lever; not part of the Philippine reference run.)

## The lesson

Two levers, two different objects — *not* two views of one build-out. The model's own first-order condition exposes the mislabeling that partial-equilibrium intuition ("capital-intensive  $\Rightarrow$  pulls capital in") would smuggle into the results; the discipline forbids stacking them.

## Health — emissions to demography and labour



- real inputs: the Global Burden of Disease ambient-PM<sub>2.5</sub> burden (43,951 attributable deaths, Philippines 2023) with real age profiles;  $M$  = energy mass-share  $\times$  concentration-response elasticity — the  $\times M$  is what separates -279 from the naive -3,400
- **mortality**: PM<sub>2.5</sub> deaths skew elderly  $\rightarrow$  saved lives are mostly retirees  $\rightarrow$  long-run output  $\approx 0$
- **morbidity**: working-age disability falls  $\rightarrow$  effective labour rises — *the* macro gain (+0.19% near-term;  $Y_{ss} = -0.003\%$ )

The signature: a small negative steady state, a larger positive transition — endogenous demography from an emissions signal, invisible to a static core. Remaining assumptions: linearity of the mapping, the morbidity elasticity.

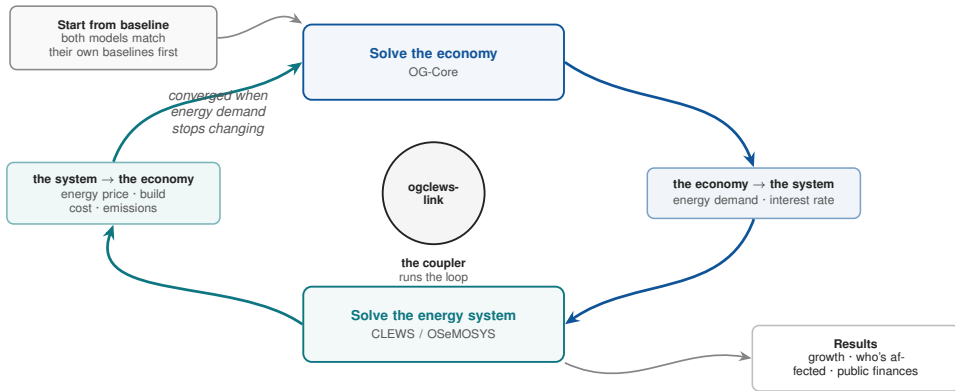
## What the economy sends back: its return, its demand

Two channels run *after* the economy solves and write ready-to-use CLEWS inputs:

- **discount rate:** the economy's equilibrium return  $r_p$  becomes the energy system's planning discount rate — so energy planning uses the *model-consistent* cost of capital. A lower return favours capital-heavy, long-lived assets.
- **energy demand:** how activity changed (industry output or consumption, with an elasticity) scales the demand the system must meet. In a single pass the change is  $\approx$  zero — latent by construction.

They move the macro economy only once the loop iterates — the quantity link of the classic coupled energy–macro models (MESSAGE-MACRO, TIMES-MACRO), but feeding an economy with households and a government instead of a single representative agent.

## Closing the loop



Re-solve the two models in turn until energy demand stops changing. Validated today: the single pass; naive price feedback can oscillate, so the loop needs damping.



**Counting each signal once**

## Five principles, each enforced by a guard

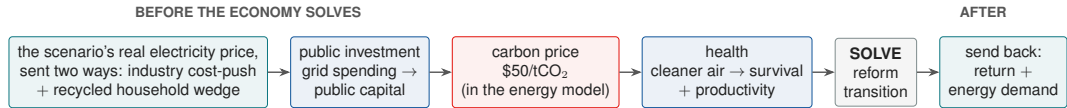
1. **One carbon price.** It enters once — a CLEWS penalty and an OG-CORE wedge on the same path — never *also* read back from the energy model's implied carbon cost.
2. **One view of an energy-cost shock.** A price wedge, a carbon wedge, and a productivity haircut are alternative doors for the *same* cost — never stacked.
3. **One view of generation capital.** A cost-push *or* an investment credit — and the capital share ( $\gamma_m$ ) is *not* a third interchangeable view: it is a factor-income lever that sheds capital (the worked example).
4. **Revenue must be routed.** A  $\tau^c$  wedge mechanically raises consumption-tax revenue; recycle it (transfers, public investment) or finance it explicitly — else a hidden fiscal expansion rides along.
5. **Public and private capital are distinct.**  $\alpha_I \rightarrow K_g$  and the firm's own  $K$  touch different objects; coupling both is complementary.

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**Does it behave? The Philippine coupled run**



## Anatomy of the flagship: the coupled run

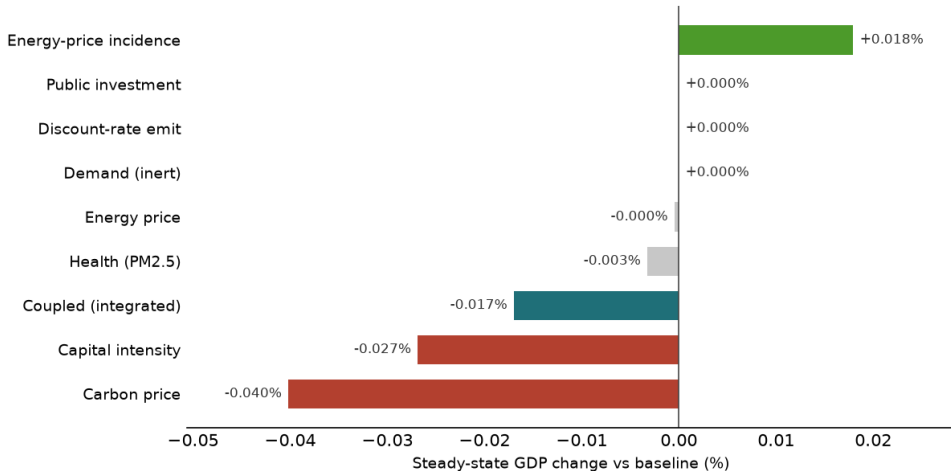


- the Philippine electricity price is near-flat  $\Rightarrow$  the energy channels contribute almost nothing (electricity output moves  $-0.18\%$ ); **the macro response is carried by health**
- here the carbon price acts in the energy model (one-price principle); the  $-0.040\%$  shown earlier is the standalone household-wedge version

| % change from baseline | steady state | $t_0$    | $t_{10}$ |
|------------------------|--------------|----------|----------|
| output $Y$             | $-0.017$     | $+0.165$ | $+0.395$ |
| consumption $C$        | $+0.008$     | $+0.195$ | $+0.467$ |

## PHL OG×CLEWS — steady-state macro impact by channel

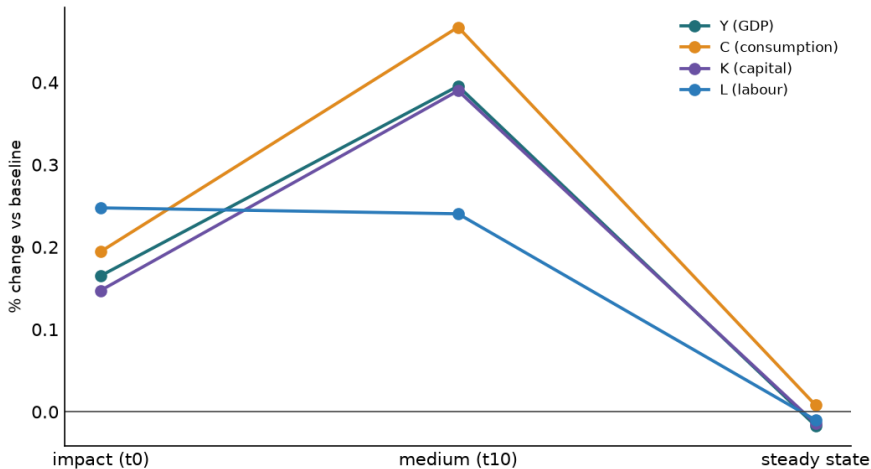
M=8 · ogcore 0.16.3 · real CLEWS 'auto' electricity price (no illustrative +20%)



Per-channel steady-state output impact, each channel run alone on the Philippine calibration. The energy price is  $\approx 0$  on the near-flat real price; the carbon price ( $-0.040\%$ ) is the clearest contraction; capital intensity is  $-0.027\%$  in aggregate (but reshapes the energy sector); health is  $-0.003\%$  at steady state yet  $+0.19\%$  near-term.

## PHL coupled result over the transition

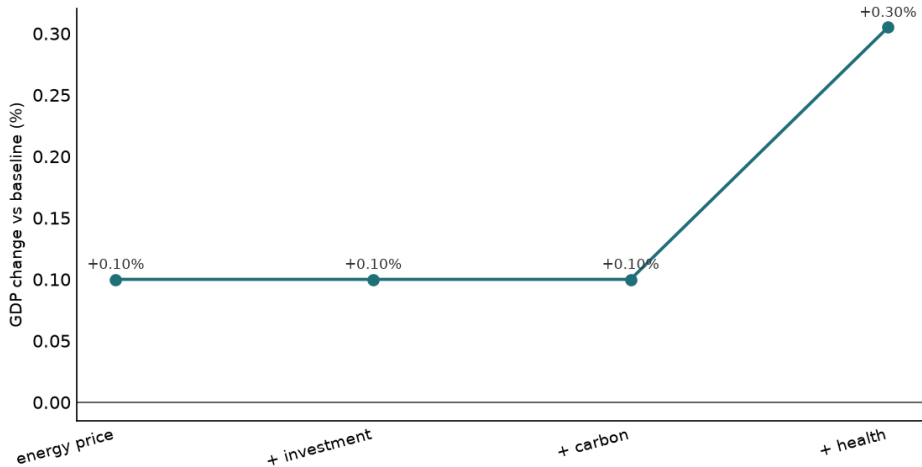
real CLEWS energy + public investment + carbon + GBD health · M=8 · ogcore 0.16.3



The coupled transition: steady state  $-0.017\%$ , but  $+0.165\%$  at  $t_0$  rising to  $+0.395\%$  by  $t_{10}$  — the working-age morbidity gain along the path, added retirees in the long run. The horizon split is the health channel's signature, and it is invisible to a static model.

## Cumulative channel contributions to the real coupled result

each step adds one channel; top step = coupled · M=8 · real CLEWS energy



Cumulative decomposition: channels layered so the top layer *equals* the coupled run — every effect attributable to a named mechanism. That auditability is the point of the channel architecture.

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**Status**

## Where the coupling stands

### Validated today

- eight channels, each one signal → one wedge, guard-railed
- single CLEWS→OG-CORE pass on *real* inputs: the scenario's real electricity price, the real Global Burden of Disease data
- each model runs in its own software environment; the link passes a few auditable files between them
- every number locked in a reference record and re-checked by tests; signs and mechanics as theory predicts

### Open, in order

- **loop closure** — the emitters are ready; the outer re-solve-and-iterate is not wired
- **the unit/deflator bridge** — carbon and investment *levels* are illustrative until it is calibrated
- **Route C** — energy as a priced production input: the structural destination

**One line:** A small, auditable library of economic mechanisms — each signal counted once, each effect attributable — rather than one opaque coupling.